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FIBER REINFORCED PALLETS

Abstract

A reinforced plastic pallet includes a pallet deck having a lower deck surface and an opposing upper deck surface, and the upper deck surface is configured for contacting goods. The reinforced plastic pallet includes one or more pallet legs extending between the lower deck surface and a base layer, wherein the base layer is reinforced with a fiber layer. A method for manufacturing a fiber-reinforced pallet having a thermoplastic part includes programming a controller with a pallet assembly profile; moving the thermoplastic part relative to a heat source to form a melted surface on the thermoplastic part; and depositing a fiber-reinforced material onto the melted surface according to the pallet assembly profile. The method may include entangling the fiber-reinforced material with the melted surface; and cooling the melted surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Application No. 63/334,369, filed Apr. 25, 2022, and the contents of which are incorporated herein by reference in their entireties for all purposes.

FIELD

[0002] The present disclosure relates to pallets for the transportation and/or handling of goods.

BACKGROUND

[0003] A pallet is used to support, store, and/or transport goods in bulk. It is known to construct pallets from various materials, such as wood or plastic. Pallets constructed from plastic may provide desirable impact performance and durability compared to conventional wooden pallets, but other mechanical properties of wooden pallets (such as, e.g., resistance to deformation) are challenging to replicate, leading to increased material usage, increased weight, increased manufacturing complexity, and or potential corrosion.

SUMMARY OF THE INVENTION

[0004] Aspects of the present invention are directed to pallet systems.

[0005] In accordance with one aspect of the present invention, a reinforced plastic pallet is disclosed. The pallet comprises a pallet deck having a lower deck surface and an opposing upper deck surface. The upper deck surface is configured for contacting goods. The pallet includes one or more pallet legs extending between the lower deck surface and a base layer, wherein the base layer is reinforced with a fiber layer.

[0006] In accordance with another aspect of the present invention, a method for manufacturing a fiber-reinforced pallet having a thermoplastic part is disclosed. The method comprises programming the controller with a pallet assembly profile; moving the thermoplastic part relative to a heat source to form a melted surface on the thermoplastic part; and depositing a continuous fiber-reinforced material onto the melted surface according to the pallet assembly profile.

[0007] In accordance with another aspect of the present invention, a method for manufacturing a fiber-reinforced pallet having a thermoplastic part is disclosed. The method comprises the steps of placing a thermoplastic part in proximity to a pellet extruder, the pellet extruder having an extruder head; loading pellets comprising thermoplastic material into the pellet extruder; heating the pellets in the pellet extruder until at least some of the pellets become plasticized material; moving the extruder head relative to the thermoplastic part so that the extruder head faces a predetermined target point on a surface of the thermoplastic part; advancing a portion of the plasticized material out of the extruder head and directly onto the surface of the thermoplastic part at the predetermined target point; and cooling said first portion of the plasticized material to permanently fuse it to the surface of the thermoplastic part at the predetermined target point.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0008] The invention is best understood from the following detailed description when read in connection with the accompanying drawings, with like elements having the same reference numerals. When a plurality of similar elements are present, a single reference numeral may be assigned to the plurality of similar elements with a small letter designation referring to specific elements. When referring to the elements collectively or to a non-specific one or more of the elements, the small letter designation may be dropped. This emphasizes that according to common practice, the various features of the drawings are not drawn to scale unless otherwise indicated. On the contrary, the dimensions of the various features may be expanded or reduced for clarity.

Included in the drawings are the following figures:

[0009] FIG. 1 depicts a fiber-reinforced pallet according to one example;

[0010] FIG. 2 is an exploded view of the pallet of FIG. 1;

[0011] FIG. 3A depicts a pallet according to another example;

[0012] FIG. 3B is a partial cross-section view of the pallet of FIG. 3A, showing a location of a fiber layer;

[0013] FIG. 4A depicts a pallet according to yet another example;

[0014] FIG. 4B is a partial cross-section view of the pallet of FIG. 4A, showing another location of the fiber layer; and

[0015] FIGS. 5 and 6 each illustrate an exemplary method of manufacturing a fiber-reinforced pallet.

DETAILED DESCRIPTION OF THE INVENTION

[0016] Aspects of the invention are described herein with reference to handling (e.g. storing, transporting, etc.) goods, such as food, pharmaceutical products, and other items known in the art. It will be understood by one of ordinary skill in the art that the example pallets described herein may be used to transport a variety of goods, and are not limited to any goods or product disclosed herein. Further, it will be appreciated that throughout this specification, the term “pallet floor” or “floor” is to be broadly construed as a surface configured to be in contact with the pallet, such as the floor of an industrial or working environment (e.g. warehouse), a level of a pallet racking system (e.g. shelves or racks), a vehicle surface (i.e. vehicle is broadly defined as any machine used for transporting goods from one location to another), or a combination thereof.

[0017] With reference to the drawings, FIG. 1 shows an exemplary reinforced plastic pallet **100**. In a non-limiting example, the plastic pallet **100** is adapted for use in handling or transporting sensitive goods requiring antimicrobial or sanitary materials, such as food. Generally, plastic pallet **100** comprises a pallet deck **102**, a base layer **120**, a fiber layer **106**, and one or more pallet legs **108**. Individual components of pallet **100** will now be described in detail below.

[0018] As shown in FIGS. 1 and 2, the pallet deck **102** may comprise an upper deck surface **110** and a lower deck surface **112**. The upper deck surface **110** is configured for contacting the goods. In a non-limiting example, the upper deck surface **110** and the lower deck surface **112** includes a generally flat, planar surface having a plurality of holes extending through an entire thickness of the pallet deck **102**. The plurality of holes desirably reduces the surface area of the upper deck surface **110** and lower deck surface **112**, increases air circulation for goods in contact with the upper deck surface **110**, and reduces the overall weight of the pallet **100**.

[0019] Further, the pallet deck **102** may have any geometry known, including a generally circular, oval, square or rectangular geometry. In certain embodiments, the perimeter shape of the upper deck surface **110** substantially matches that of the lower deck surface **112**. Still further, the pallet deck **102** may be molded from thermoplastic or other polymer materials, such as high-density polyethylene (HDPE), polypropylene (PP), etc. The upper deck surface **110** may comprise polymer

material that is different from that of the lower deck surface **112**.

[0020] The pallet deck **102** according to the present disclosure may be formed using conventional or customized computer numerical control (CNC) machines that use data to control and monitor movement and/or activation of machine parts. In an exemplary embodiment, the CNC machine has an extruder. The extruder can be any conventional extruder that extrudes material. For example, the extruder can have a rotating helical screw or a reciprocating plunger. In some aspects, the pellet extruder is part of a large format 3D printer (e.g. a Cartesian printer).

[0021] The CNC machine also has a controller that works with a number of motors and drive components in the machine. The motors and drive components control movement of a carrier unit that moves the extruder relative to an article. In addition, the motors and drive units control operation of the extruder. The controller operates each motor to execute programmed motions and functions. One or more sensors send feedback to the controller to monitor the position or condition of the drive units and extruder.

[0022] While the present disclosure is described with reference to use of a thermoplastic or other polymer materials as an extruded material, the invention is not so limited. A person of ordinary skill in the art will understand from this disclosure, however, that other extruded materials are within the scope of this invention. The extruded materials compatible with the invention as disclosed herein include all materials which made be made molten and extruded through an extruder including, e.g., plastic materials, plasticized materials, metallic materials, ceramic materials, conductive materials, and adhesives. The feedstock for these materials may be supplied to the extruder in any suitable form, such as, for example, pellet, filament, resin, powder, and wire form.

[0023] In operation, the CNC machine has an extruder head and nozzle tip that can be positioned and oriented relative to a target surface. The controller is configured to control operation of motors and receive feedback from sensors to monitor the drive units and extruder. After positioning the extruder head and nozzle tip over the target surface, the controller activates the extruder to extrude material (e.g. thermoplastic materials), which can be repeated according to a predetermined program, pattern or shape, until pallet deck **102** is formed.

[0024] Additionally or optionally, the pallet deck **102** may comprise customized changes or refinements, including but not limited to attachment of one or more types of material to change the surface design or geometry of a portion or an entirety of pallet deck **102**. Apparatuses and processes for modification of pallet deck **102**, including use of conventional or customized computer numerical control (CNC) machines and automatic processes (e.g., 3D printing), may include apparatuses and processes described in U.S. application Ser. No. 17/083,639, the contents of which is incorporated by reference herein in its entirety. Additionally, or optionally, the customized changes or refinements may comprise application of one or more surface enhancements (e.g., a functional surface, ornamental surface, or indicia to the surface) to a portion or an entirety of pallet deck **102**. Apparatuses and processes for application of surface enhancement(s) to pallet deck **102** may include apparatuses and processes described in U.S. Provisional Application Ser. No. 62/977,963, the contents of which is incorporated by reference herein in its entirety.

[0025] One or more pallet legs **108** extend between the lower deck surface **112** and the base layer **120**. The base layer **120** may comprise pallet legs **108**. In a non-limiting example, the pallet legs **108** are configured to have a height sufficient to elevate at least the upper deck surface **110** from a pallet floor. In the exemplary embodiment shown in FIGS. **1** and **2**, pallet legs **108** are spaced apart from each other and may be positioned along perimeter edges defined by the upper deck surface **110** and/or lower deck surface **112**. Alternative embodiments may include one or more pallet legs **108** located centrally between the lower deck surface **112** and the base layer **104**. Lower deck surface **112** and base layer **120** may alternatively be joined through a single pallet leg which frames the entire perimeter of the pallet (e.g., forming the sides of a rectangular prism).

[0026] The pallet legs **108** may comprise the same or different thermoplastic or polymer material

compared to that of the pallet deck **102**.

[0027] Additionally or optionally, the base layer **120** may comprise a plurality of stringers **104**. In an exemplary embodiment, as shown in FIG. 2, attached to a floor-contacting portion of the one or more pallet legs **108** are the plurality of stringers **104**. The plurality of stringers **104** may be provided between the two adjacent pallet legs **108**, such that the plurality of stringers **104** generally extend along an outer perimeter of the pallet **100**, as defined for example, by the perimeter edges of the upper deck surface **110** and/or lower deck surface **112**. Additionally or optionally, the plurality of stringers **104** may be provided along a central region of the pallet deck **102** when one or more adjacent pallet legs **108** are also provided along the central region of the pallet deck **102**. The plurality of stringers **104** may provide improved impact strength, stronger structural properties, or other improved reinforcement qualities to the pallet **100**. Further, the plurality of stringers and the pallet legs **108** may be connected, such that the pallet legs **108** are spaced apart and the separation allows for the tines of a pallet jack or forklift to be inserted therebetween, for example.

[0028] Turning now to FIGS. 3A-3B, a fiber layer **106** may be integrated within the base layer **120**, the base layer **120** comprising the plurality of stringers **104**, for providing improved reinforcement qualities to the pallet **100**. Improved reinforcement qualities may be qualitatively or quantitatively measured, for example, by a reduction of a flexural displacement of the upper deck surface **110** when the upper deck surface **110** is in contact with the goods. Specifically, flexural displacement of pallet **100**, particularly of components subject to high tension, such as the upper deck surface **110**, may be reduced when the pallet **100** is subjected to edge racking conditions. One skilled in the art would understand from the description herein that edge racking conditions include a pallet racking system typically involving racks that support pallets by only two edges and provide no central support.

[0029] The fiber layer **106** may comprise continuous fiber-reinforced materials, such as a fiber-reinforced composite tape, a fiber-reinforced composite sheet, or combinations thereof. In an exemplary embodiment, the fiber layer **106** may comprise continuous fiber reinforced thermoplastic (CFRTP) tape, with fiber including carbon fiber material or glass fiber material, each of which can be combined with an epoxy resin (e.g. PP, PP-Black, polyethylene (PE), polyethylene terephthalate glycol (PETG), aPET-Black, PA6-Black).

[0030] Generally, the format, structure, orientation, and other material properties of continuous fiber material **106** may depend on the type and design of pallet **100**. Fiber layer **106** may be strategically placed in areas of the pallet **100** which inherently represent weak points in terms of load distribution and/or structural properties. In one non-limiting example, as best shown in FIG. 3B, the fiber layer **106** may be positioned near a lower surface (e.g., closer to the pallet floor) of the plurality of stringers **104**. Additionally or optionally, another fiber layer **106** may be integrated with the lower deck surface **112** of the pallet deck **102**. Further, another fiber layer **106** may be positioned at an interface between the plurality of stringers **104** and the respective floor-contacting portions of the one or more pallet legs **108**. Still further, another fiber layer **106** may be positioned at an interface between the lower deck surface **112** and the respective one or more pallet legs **108**. Additionally or optionally, the fiber layer **106** may be similarly integrated with portions of upper deck surface **110**.

[0031] Concerning the integration of the fiber layer **106** with at least one component of the pallet **100**, such as the base layer **120**, the base layer **120** comprising the plurality of stringers **104** and/or the one or more pallet legs **108**, the continuous fiber-reinforced materials **106** may be directly transferable to a pallet mold for the purposes of over-molding during a molding process. In one non-limiting example, continuous fiber-reinforced tape **106** may be molded directly into the pallet **100**, such that tape **106** is not a secondary assembled part or a pre-molded reinforced component that is then over-molded. Specifically, the fiber-reinforced tape **106** may be molded directly into the pallet **100** using apparatuses and processes, such as conventional or customized computer numerical control (CNC) machines and automatic processes (e.g., 3D printing), as described in

U.S. application Ser. No. 17/083,639 and U.S. Provisional Application Ser. No. 62/977,963, the contents of each of which is incorporated by reference herein in its entirety.

[0032] FIG. 5 illustrates an exemplary method of manufacturing a fiber-reinforced pallet. The method **1000** includes one or more steps including forming a melted surface on at least one component of the pallet and depositing a continuous fiber-reinforced additive material onto the melted surface. Additional details of method **1000** are set forth below with respect to the elements of pallet **100**.

[0033] In a first step **1100**, a controller is programmed with a pallet assembly profile.

[0034] In a second step **1200**, a focused infrared (IR) heater produces a heat flux adequate to melt the top skin or surface of at least one component of pallet, such as the plurality of stringers **104**. In particular, the focused IR heater produces the heat flux as the outer perimeter edges of pallet **100** (along which the plurality of stringers **104** extends) is moved relative to the IR heater. The top surface is melted to a desired depth. The depth of melting is controlled based on one or more parameters, including but not limited to intensity setting of the IR heater, the spacing between the heater and the part, and the velocity at which the part is moved relative to the heater. The entire top surface can be melted. Alternatively, only the area(s) of the top surface corresponding to the placement of the fiber layer **106** is (are) melted.

[0035] In a third step **1300**, an applicator immediately applies continuous fiber-reinforced tape or additive material onto the melted surface of the part. The continuous fiber-reinforced additive material may be in the form of particles, pellets, flakes, regrind, powders or may be sheet fed from one or more rolls. The applicator can move relative to the surface of at least one component of pallet **100** to apply additive material to different areas on the melted surface. The controller controls the movement of the applicator and/or the at least one component of pallet **100** so as to apply additive material according to the pallet assembly profile. The additive material can be a variety of thermoplastic materials, including but not limited to PP, HDPE, polycarbonate (PC), and polyamide (PA). The additive material adheres to the sticky, molten surface of the part after it is applied.

[0036] In an additional or optional fourth step **1400**, the fiber-reinforced pallet **100** exits a tunnel. The term “exit” as used herein can refer to a moving part being conveyed out of a stationary tunnel, a moving tunnel moving past a stationary part, or a moving part moving out of a moving tunnel traveling at a different speed and/or different direction than the moving part.

[0037] Additionally or optionally, one skilled in the art would understand from the description herein that to enable over-molding, several structures or techniques may be employed, including but not limited to support pins, spring pins, applied vacuum to suspend the continuous fiber-reinforced material **106** in strategic locations of the pallet **100**, or combinations thereof.

[0038] FIG. 5 depicts an example method comprising steps that are performed sequentially in the order recited. However, it should be understood from the description herein that one or more steps may be omitted and/or performed out of the described sequence of the process while still achieving the desired result.

[0039] FIG. 6 illustrates another exemplary method of manufacturing a fiber-reinforced pallet, such as method **2000**. The method **2000** is similar to the steps and details of method **1000**. For example, the method **2000** includes one or more steps including forming a melted surface on at least one component of the pallet and depositing an additive material onto the melted surface. However, method **2000** differs from method **1000** in several respects, as discussed below with respect to the elements of pallet **100**.

[0040] In a first step **2100**, a thermoplastic part is placed in proximity to a pellet extruder, the pellet extruder having an extruder head. In particular, the pallet **100** having a thermoplastic part is placed in proximity to the pellet extruder, such that the outer perimeter edges of pallet **100** (along which the base layer **120**, the base layer **120** comprising the plurality of stringers **104** extends) is moved relative to the pellet extruder.

[0041] In a second step **2200**, pellets comprising thermoplastic material are loaded into the pellet extruder. Additionally or optionally (in a subsequent step), the pellets are heated in the pellet extruder until at least some of the pellets become plasticized material.

[0042] In a third step **2300**, the extruder head is moved relative to the thermoplastic part so that the extruder head faces a predetermined target point on a surface of the thermoplastic part. Additionally or optionally (in a subsequent step), a first portion of the plasticized material is advanced out of the extruder head and directly onto the surface of the thermoplastic part at the predetermined target point. In particular, the extruder head is moved such that the extruder head faces the outer perimeter edges of pallet **100** (along which the base layer **120**, the base layer **120** comprising the plurality of stringers **104** extends). Additionally or optionally (in a subsequent step), the first portion of the plasticized material is advanced onto the entire top (or bottom) surface of the base layer **120** comprising the plurality of stringers **104**. Alternatively, the first portion of the plasticized material is advanced only onto the area(s) of the top surface corresponding to the placement of the fiber layer **106** is (are). The characteristics of the advancement of the first portion of the plasticized material is controlled based on one or more parameters, including but not limited to the spacing between the extruder head and the part, and the velocity at which the first portion of the plasticized material is deposited onto the pallet **100**.

[0043] In a fourth step **2400**, said first portion of the plasticized material is cooled to permanently fuse it to the surface of the thermoplastic part at the predetermined target point.

[0044] In a fifth step **2500**, the surface of thermoplastic part is moved relative to a heat source to form a melted surface on the thermoplastic part. This step is similar to step **1200** of method **1000**, as described above.

[0045] In a sixth step **2600**, an additive material is deposited onto the melted surface, entangled with the melted surface; and the melted surface is cooled. This step is similar to step **1300** of method **1000**, as described above. In particular, the additive material may include material capable of creating a fiber layer such as fiber-reinforced tape or other fiber-based material. The additive material can additionally or alternatively be a variety of thermoplastic materials, including but not limited to PP, HDPE, polycarbonate (PC), and polyamide (PA). The additive material adheres to the sticky, molten surface of the part after it is applied.

[0046] FIG. **6** depicts an example method comprising steps that are performed sequentially in the order recited. However, it should be understood from the description herein that one or more steps may be omitted and/or performed out of the described sequence of the process while still achieving the desired result.

[0047] Another embodiment of the pallet **100** according to the present invention is illustrated in FIGS. **4A-4B**. The components of this embodiment, such as pallet **200**, generally correspond to the components of pallet **100**. However, this embodiment differs from the first embodiment described above in some respects. In one example, the fiber layer **106** is positioned in a central region between an upper and lower surface of the plurality of stringers **104**.

[0048] Although the invention is illustrated and described herein with reference to specific embodiments, the invention is not intended to be limited to the details shown. Rather, various modifications may be made in the details within the scope and range of equivalents of the claims and without departing from the invention.

Claims

1. A reinforced plastic pallet comprising: a pallet deck having a lower deck surface and an opposing upper deck surface, the upper deck surface configured for contacting goods; one or more pallet legs extending between the lower deck surface and a base layer, wherein the base layer is reinforced with a fiber layer.

2. The reinforced plastic pallet of claim 1, wherein the fiber layer is a multilayered composite fiber

having a first fiber layer and a plurality of second fiber layers overlapped over or under the first fiber layer in a single, parallel direction.

3. The reinforced plastic pallet of claim 1, wherein the fiber layer is a multilayered composite fiber having a first fiber layer and a plurality of second fiber layers overlapped over or under the first fiber layer, and wherein each of the plurality of second fiber layers are overlapped over or under the first fiber layer at a respective angle relative to the first fiber layer.

4. The reinforced plastic pallet of claim 1, wherein the angle is between 0 and 90 degrees.

5. The reinforced plastic pallet of claim 1, wherein the fiber layer is positioned near a lower surface of the base layer.

6. The reinforced plastic pallet of claim 1, wherein the fiber layer is positioned in a central region between an upper and lower surface of the base layer.

7. The reinforced plastic pallet of claim 1, wherein the base layer comprises a plurality of stringers.

8. The reinforced plastic pallet of claim 7, wherein the fiber layer comprises one or more of a fiber-reinforced composite tape, a fiber-reinforced composite filament, and a fiber-reinforced composite sheet.

9. The reinforced plastic pallet of claim 7, wherein the fiber layer comprises one or more of an aramid fiber, a carbon fiber, a glass fiber, carbon reinforced plastic fiber, a glass reinforced plastic fiber, a basalt fiber, and a metal fiber.

10. The plastic pallet of claim 1, at least one of the upper deck surface and the lower deck surface is reinforced with one or more fiber layers.

11. The plastic pallet of claim 1, further comprising another fiber layer positioned at an interface between the base layer and the one or more pallet legs.

12. The plastic pallet of claim 1, further comprising another fiber layer positioned at an interface between the lower deck surface and the one or more pallet legs.

13. The plastic pallet of claim 1, wherein at least the upper deck surface comprises antimicrobial material.

14. A method for manufacturing a fiber-reinforced pallet having a thermoplastic part, the method comprising the steps of: A. programming a controller with a pallet assembly profile; B. moving the thermoplastic part relative to a heat source to form a melted surface on the thermoplastic part; C. depositing a fiber layer material onto the melted surface according to the pallet assembly profile.

15. The method of claim 11, further comprising the steps of: D. entangling the fiber-reinforced material with the melted surface; and E. cooling the melted surface.

16. The method of claim 14, wherein the step of moving the thermoplastic part relative to the heat source comprises moving the thermoplastic part at different speeds and/or different directions.

17. The method of claim 14, further comprising the step of applying heat to the additive material and melted surface after the step of depositing the additive material onto the melted surface.

18. The method of claim 14, wherein the fiber layer materials comprise one or more of an aramid fiber, a carbon fiber, a glass fiber, carbon reinforced plastic fiber, a glass reinforced plastic fiber, a basalt fiber, and a metal fiber.

19. The method of claim 18, further comprising the step of flattening and mating the additive material with the melted surface.

20. The method of claim 14, wherein the step of depositing an additive material onto the melted surface comprises moving an applicator containing the additive material over the melted surface.

21. The method of claim 20, wherein the applicator is part of a CNC machine having a controller.

22. The method of claim 21, wherein the CNC machine is a Cartesian printer.

23. The method of claim 22, wherein the CNC machine is a CNC controlled robotic arm.

24. A method for manufacturing a fiber-reinforced pallet having a thermoplastic part, the method comprising the steps of: A. placing a thermoplastic part in proximity to a pellet extruder, the pellet extruder having an extruder head; B. loading pellets comprising thermoplastic material into the pellet extruder; C. heating the pellets in the pellet extruder until at least some of the pellets become

plasticized material; D. moving the extruder head relative to the thermoplastic part so that the extruder head faces a predetermined target point on a surface of the thermoplastic part; E. advancing a first portion of the plasticized material out of the extruder head and directly onto the surface of the thermoplastic part at the predetermined target point; and F. cooling said first portion of the plasticized material to permanently fuse it to the surface of the thermoplastic part at the predetermined target point.

25. The method of claim 24, further comprising the steps of: G. moving the surface of thermoplastic part relative to a heat source to form a melted surface on the thermoplastic part; H. depositing an additive material onto the melted surface; I. entangling the additive material with the melted surface; and J. cooling the melted surface.

26. The method of claim 25, wherein the plasticized material and the additive material form a functional surface on the surface of the thermoplastic part.

27. The method of claim 26, wherein the surface of the thermoplastic part comprises a lower deck surface of a pallet deck.

28. The method of claim 25, wherein the plasticized material and the additive material forms an ornamental surface on the surface of the thermoplastic part.

29. The method of claim 24, wherein steps D, E and F are repeated at a plurality of predetermined target points on the surface of the thermoplastic part.

30. The method of claim 25, wherein steps G, H, I, and J are repeated at a plurality of predetermined target points on the surface of the thermoplastic part.

31. The method of claim 24, wherein the pellets are formed of the same material as the thermoplastic part.

32. The method of claim 24, wherein the pellet extruder is part of a CNC machine having a controller.

33. The method of claim 24, wherein the controller controls movement of the extruder head.

34. The method of claim 33, further comprising the step of programming the controller to move the extruder head to one or more positions relative to the thermoplastic part.

35. The method of claim 24, wherein the pellet extruder is connected to a large format 3D printer.

36. The method of claim 35, herein the 3D printer is a Cartesian printer.
